PUJIT BALANTHIRAN

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PROFESSIONAL SUMMARY

Google TechSprint Top-10 Finalist and full-stack software engineer with 3 deployed, production-grade systems spanning AI pipelines, real-time data processing, graph databases, and cloud-native architecture. Proficient in Java, Python, C++, JavaScript, TypeScript, and React.js with a strong foundation in Data Structures & Algorithms, OOP, DBMS, REST API design, and Agile software engineering. Experienced with Docker, CI/CD via GitHub Actions, and cloud deployment on Google Cloud Run and Firebase. Seeking SWE internship roles in full-stack or backend engineering.

TECHNICAL SKILLS

Languages:	Java, Python, C++, JavaScript, TypeScript, HTML5, CSS3
Frameworks & Libraries:	React.js (v18/19), Node.js, Express.js, Tailwind CSS, Framer Motion, D3.js, Recharts, Leaflet.js, Jest
Databases:	MongoDB Atlas, Neo4j (Graph DB), SQL, Firebase Realtime DB, Supabase
Cloud & DevOps:	Google Cloud Run, Google Cloud Functions, Firebase, Docker, GitHub Actions CI/CD, Microservices
AI & APIs:	Google Gemini Pro, Google Speech-to-Text API, WhatsApp Cloud API, REST API design
Methodologies:	Agile / Scrum, Data Structures & Algorithms, OOP, DBMS, Software Engineering, Cryptography
Tools:	Git, Vite, JWT, bcryptjs, Multer, Helmet, Postman, MATLAB, RStudio, Vivado

PROJECTS

NeedGraph— AI-Powered Crisis Prediction & Volunteer Coordination Platform

github.com/pujit23/needgraph_final

React· Vite· Tailwind CSS· D3.js· Leaflet.js· Node.js· Google Cloud Functions· Google Gemini Pro· Google Speech-to-Text· Neo4j· WhatsApp Cloud API· Google Cloud Run· GitHub Actions CI/CD· Docker

- Engineered a real-time AI ingestion pipeline processing WhatsApp voice notes in **5+ regional languages** via Google Speech-to-Text and Gemini Pro, extracting severity scores (0–100), need classification, affected population, and geo-coordinates— delivering actionable intelligence in **under 60 seconds** end-to-end.
- Architected a Neo4j graph database model mapping **1,000+ interconnected social-risk nodes**; implemented a cascade prediction algorithm detecting multi-level crisis chains (food insecurity→ school dropout→ mental health crisis) and triggering automated coordinator alerts **up to 3 weeks** before escalation.
- Built and deployed a **13-page low-latency React dashboard** with D3.js network graph explorer, Leaflet.js real-time need map, crisis simulator, and live volunteer dispatch on Google Cloud Run with automated CI/CD— architected to support coordination across **10,000+ NGOs** across India.

CycHigh— Full-Stack Cycling Encyclopedia & Community Platform

github.com/pujit23/cychigh

React 18· Vite· Tailwind CSS· Framer Motion· Recharts· Leaflet.js· Node.js· Express.js· MongoDB Atlas· Supabase· JWT· bcryptjs· Multer·Docker

- Designed a scalable full-stack web application backed by a MongoDB Atlas database of **120 bicycles across 25 brands** with complete technical specifications, INR pricing, maintenance guides, and upgrade recommendations— serving **50+ REST API endpoints**.
- Implemented an intelligent **5-parameter recommendation algorithm** scoring user inputs across skill level, terrain, budget, and riding purpose to surface top-5 matched cycles with percentage scores; engineered a side-by-side comparison engine with automated value-highlighting and client-side PDF export.
- Built a secure Node.js/Express.js backend with JWT auth, bcryptjs hashing, Multer image upload, rate limiting (Helmet), and a full admin dashboard with CRUD and **bulk JSON import for 120+ records**; integrated Supabase real-time subscriptions for live ride tracking, leaderboards, and community chat.

FindSpace— Secure Campus Peer-to-Peer Marketplace
github.com/pujit23/Findspace
React 19 · React Router DOM v7 · Vite · TypeScript · Vanilla CSS · Lucide React · Firebase Authentication

- Developed a secure, identity-verified peer-to-peer trading platform using React 19 and Firebase Authentication, enforcing campus-exclusive access through OTP-based phone verification — achieving **zero unauthorized transactions** across **8+ application pages**.
- Engineered a multi-context state management architecture using React Context API (AuthProvider, CartProvider, ProductProvider, ThemeContext) with full **TypeScript type safety**, managing auth state, cart logic, real-time product listings, and theme switching.
- Delivered a fully custom mobile-responsive CSS design system leveraging CSS variables, glassmorphism, dynamic gradients, and micro-animations with **ARIA-compliant** focus trapping across all modal components— achieving **100% mobile responsiveness**.

HACKATHONS & COMPETITIONS

Google TechSprint— Top 10 Finalist

• Ranked in the **top 10 out of all participating teams**; built and presented a fully functional full-stack solution with real-time data features under strict time constraints, evaluated live by Google engineers.

Google Solutions Challenge— Participant

• Developed a socially impactful platform addressing **UN SDG Goal 3 (Good Health)** using React, Firebase, and Google Cloud Run— delivered a working deployed prototype to Google developer mentors.

Smart India Hackathon (SIH)— Participant

• Proposed and prototyped a full-stack engineering solution for a real government problem statement at **India's largest national-level hackathon**— Node.js REST API backend with a React frontend.

Google DevSprint— Participant

• Shipped a working Google Cloud API integration within a **24-hour build window** in a fast-paced engineering sprint organised by the Google Developer ecosystem.

EDUCATION

Vellore Institute of Technology– AP University 2024– 2028
B.Tech in Computer Science & Engineering
Relevant Coursework: Data Structures & Algorithms, OOP, Digital Image Processing, Assembly Language Programming, Software Engineering, DBMS, Cryptography

CERTIFICATIONS

HackerRank SQL— Basic & Intermediate — Verified SQL proficiency: joins, aggregations, subqueries, window functions

SQL for Data Science — Kaggle— applied SQL for structured data analysis

MATLAB Onramp — MathWorks— signal processing, matrix operations, and image processing fundamentals

LEADERSHIP & COMMUNITY

Microsoft Student Chapter, VIT-AP— Active Member

• Technical workshops, peer learning, and collaborative engineering projects within the chapter community.

Google Developer Group (GDG), VIT-AP— Active Member

• Participated in Google-ecosystem events, tech talks, and hands-on coding sessions; contributed to **3+ collaborative build sprints** within the developer community.

Computer Society of India (CSI), VIT-AP— Member

• Engaged in technical seminars, competitive programming events, and CS community activities.